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Alice has 2 integers L and R. Let X be the number we get by xoring all numbers in the range $[L, R]$ (including L and R). He recently learnt about even and odd numbers and wonders what will be the parity on X.

Input Format:

First line contains 2 integers L and R

Output Format:

Print **"even"** if the number X is even else print **"odd"** without the quotes.

Constraints:

$1 \leq L \leq R \leq 10^{18}$

Sample Input	Sample Output
3 5	even

Time Limit: 1

Memory Limit: 256

Source Limit:

Explanation

Here L is 3 and R is 5

Now X=(3^4^5) where '^' represents bitwise xor.

X=2.

Since X is even, the output is "even".

Enter your code or Upload your code as file. Save C++14 (g++ 10.3.0)

```
long long xorUpto(long long n) {
    if (n % 4 == 0) return n;
    if (n % 4 == 1) return 1;
    if (n % 4 == 2) return n + 1;
    return 0;
}

int main() {
    long long L, R;
    cin >> L >> R;

    long long X = xorUpto(R) ^ xorUpto(L - 1);

    if (X % 2 == 0)
        cout << "even";
    else
        cout << "odd";

    return 0;
```

23:2 vscode

Test against custom input ▾ Compile & Test code Submit code

Submission ID: 127930980

RESULT: Accepted Refer judge environment